

GuardSynth: Verified Constraint Rendering for Visual Action Learning

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Abstract

We study whether a source-bearing executable constraint representation can preserve previously observed benefits of supplied natural-language constraints in visual language model action selection. Our pipeline separates supplied assumptions, typed constraint lowering, bounded solver checks, controlled-language rendering, and visual model learning. A controlled temporal task compares an unconstrained chain of cognition (P0), a supplied natural-language constraint (P1), and an EBLC-Core-verified rendering of the same constraint (P2). Constraints are visible at training and inference in both constrained conditions. Nine local LoRA fits completed 648 optimizer updates and 2,592 action predictions. On the standard benchmark, P0 violation/safe completion was 0.319/0.681; both P1 and P2 achieved 0.000/1.000 across all three seeds. A new generator-seed sample reproduced this pattern. The prospectively fixed point criteria support preservation within this small reused template family, not formal noninferiority or road generalization. Follow-up clearance-time changes increase P2 violation to 0.167–0.250, revealing limited robustness. Separately, layered checks detect all 144 supported injected constraint errors but miss 24 consistently corrupted-source controls; verification’s contribution is scoped constraint quality, not an additional causal performance gain.

1 Question and scope

Verification of a formal representation does not establish the truth of its source assumptions or the behavior of a learned policy. We therefore distinguish semantic correspondence, behavioral effect preservation, and real-scene source applicability. The present experiment tests the first two under supplied synthetic assumptions. Existing reviewed road scenes provide a separate application case study, not an independent road-safety test.

2 Attributed prior evidence

Project01’s temporal guard experiment trained Qwen3-VL-2B LoRA adapters for three seeds. Its reported standard-task mean violation rates were 0.319 for original CoC and 0.000 for inline natural-language constraints; safe goal completion was 0.681 and 1.000. These are prior Project01 results, not current P2 measurements. On unseen time values, safe completion was 0.625 and 0.649, indicating limited numerical generalization. We reuse that project’s generator, visual processor, and optimizer harness without modifying its source or results. The historical task contains deliberately supplied guards rather than automatically extracted road constraints. Its earlier inline-NL condition remains P1’s reference; it is never relabeled as a previously demonstrated P2 result.

3 Method

The restricted supplied temporal profile identifies ego, a pedestrian conflict zone, a no-entry obligation, and a continuous-clear release duration. The world assumes exactly one occupied-to-clear transition without reoccupation. Lowering uses integer millisecond tick counts and an implication from entry to entry time being no earlier than clearance plus the duration. The shared EBLC Core parser checks typing and the SMT compiler checks feasibility and boundary cases. The renderer reads only validated contract fields; candidate labels and action answers are not inputs to constraint generation. This is a controlled profile, not a claim that the general EBLC lifecycle or real-road extraction is solved. The modeled decision is entry admissibility: candidate waiting periods are fixed by the controlled task. The Core does not model velocity during a wait; the CNL comparison does not certify kinematic equivalence outside this candidate family.

4 Frozen experiment

Protocol `paper1-method-preservation-v0.1` fixes seeds 42, 17, and 123; 192 training scenes, two counterfactual contracts per scene; three epochs, batch size two, accumulation eight, and 72 optimizer updates per arm and seed. All arms use the same Qwen checkpoint, visual inputs, targets, and optimizer. P1 and P2 append their constraint to the original CoC during both learning and inference. Only assistant answer tokens receive loss. P0 deliberately lacks the contract needed to distinguish counterfactual pairs.

Each checkpoint is evaluated on the historical benchmark distribution, a new fixed generator-seed sample, and unseen temporal values. The independent program scorer uses environment times and candidate progress, not generated CNL or model self-assessment. We report violations, safe goal completion, deadlock as an unnecessary-stop proxy, accuracy, coverage, paired contract accuracy, and raw per-example outputs. Template-cluster paired bootstrap intervals describe uncertainty conditional on this small task family.

Before new outcomes, we fixed per-seed level criteria: P1 and P2 violation at most 5% and safe completion at least 95%; P2 improvement over P0 at least 20 percentage points on both violation and safe completion; and no more than 5 percentage points degradation from P1 on either endpoint. The five-percentage-point tolerance is a prospectively chosen operational criterion, not a margin supplied by the historical study. These operational criteria do not establish formal noninferiority. Equally weak P1 and P2 fail.

5 Results

All nine fits completed 72 updates each and saved distinct adapters. Each adapter contains 224 tensors, including 112 nonzero LoRA B matrices. All 2,592 predictions were valid. Table 1 reports the new Project04 measurements; historical values were not copied into these cells. P0/P1 deliberately reuse the prior generator, seeds, and task inputs as replication controls, not independent new training cohorts. P2 uses a different rendered constraint string.

Every seed met the prespecified P1 replication, P2 level, P2–P0 improvement, and P2–P1 preservation point criteria on both standard evaluations. The new-seed P2–P0 violation difference was -0.3160 , with conditional template-cluster 95% bootstrap interval $[-0.4348, -0.1832]$; safe completion increased by the same magnitude. P2–P1 differences were zero, with degenerate empirical intervals $[0, 0]$. These intervals describe the observed templates and do not prove a population noninferiority margin. In particular, the interval does not establish a minimum 20-percentage-point

All six distinct storyboard templates in the reused controlled task

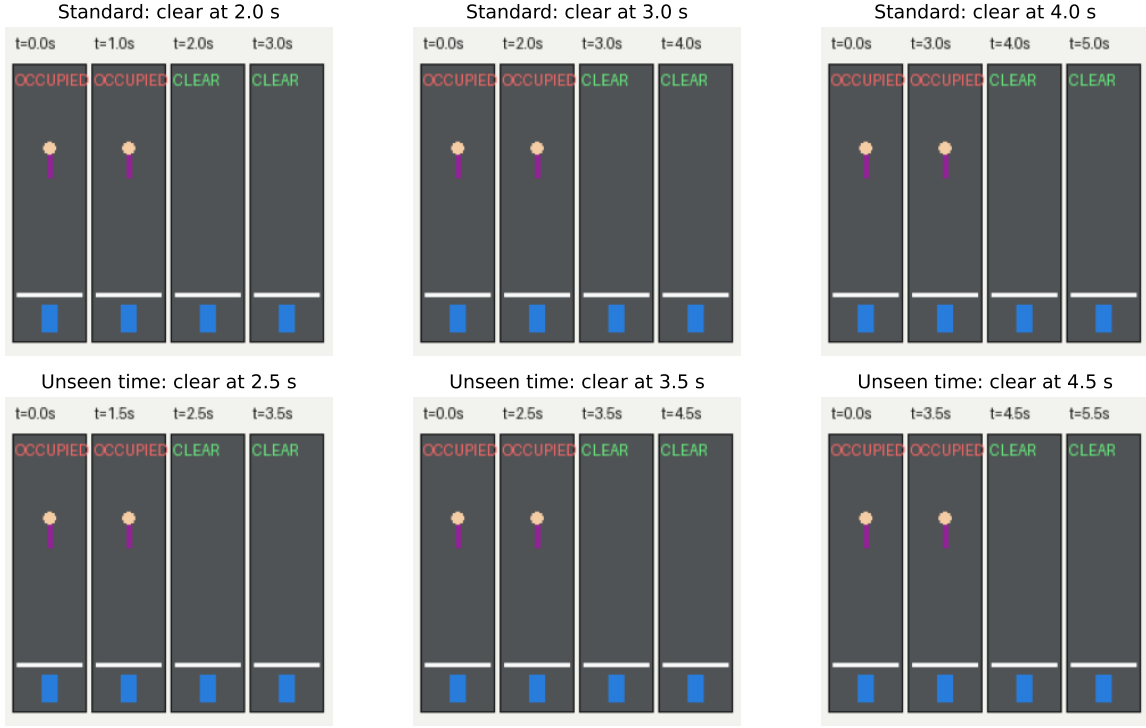


Figure 1: All six distinct rendered storyboards in the reused Project01 task. Each contains four temporal panels. Repeated generator seeds change sample frequencies and candidate label rotations, not this visual environment family.

improvement over broader environments.

The unseen-time results are descriptive secondary evidence: P1 safe completion remained limited at .6389, whereas P2 reached 1.0000 with accuracy .8993. The gap between completion and accuracy reflects safe but suboptimal later entries. Contract-pair accuracy was .0000/.2778/.7986 for P0/P1/P2 on unseen time, versus .0000/1.0000/1.0000 on the standard evaluations. This does not isolate a solver benefit or establish general numerical reasoning.

The bridge passes 100 SMT queries covering four release durations and boundary/holding cases. The supplied contract bridge also rejects unsupported subject, unit, obligation, and duration changes. A deleted-entry-gate mutation is detected by the boundary queries. The paired data export has 2,016 records across seed/split combinations; these are not 2,016 independent visual scenes.

6 Follow-up diversity and constraint quality

The D1–D4 follow-up was designed after the primary results and frozen separately before new predictions. It reuses all nine adapters without training on the new diagnostic cases. D1 separately varies clearance time, duration, their combination, and candidate boundaries, with 24 balanced rows per stratum. Displayed and tested boundaries differ by 0.1 seconds, avoiding hidden precision requirements. D2 uses two fixed, field-checked paraphrases plus canonical wording. D3 holds text and candidates fixed while retaining, blanking, or swapping the storyboard. The swap shifts visual

Evaluation	Arm	Violation	Safe completion	Accuracy
Benchmark	P0	.3194	.6806	.5000
	P1	.0000	1.0000	1.0000
	P2	.0000	1.0000	1.0000
New seed	P0	.3160	.6840	.5000
	P1	.0000	1.0000	1.0000
	P2	.0000	1.0000	1.0000
Unseen time	P0	.3750	.6250	.5000
	P1	.3611	.6389	.6250
	P2	.0000	1.0000	.8993

Table 1: Three-seed means from newly executed matched fits. Deadlock was zero and coverage was one for all arms/splits. New seeds reuse the same templates.

clearance by one second; retained-source and image-implied references are evaluated separately. Missing and contradictory visual inputs are diagnostics, not road error rates. Identical P0 inputs are cached rather than counted as independent observations.

The completed follow-up contains 2,160 scored rows from 1,476 actual model calls. D1 has nine distinct images, six absent from the original six-image experiment; each stratum has only three clearance-time groups, with label rotations and contract pairs kept together in the paired seed/group bootstrap.

Diagnostic	P1 viol.	P2 viol.	P1 safe	P2 safe
Clear time only	.3750	.1667	.6250	.8194
Duration only	.0278	.0000	.9722	1.0000
Joint change	.4028	.2500	.5972	.7222
Boundary candidates	.3889	.1528	.6111	.8472
Paraphrase 1	.0556	.0000	.9444	1.0000
Paraphrase 2	.1250	.0417	.8750	.9583

Table 2: Exploratory robustness results using the frozen original adapters. All conditions have full output coverage. These failures limit the standard-task preservation finding; correct constraints do not guarantee model adherence.

P2’s joint-change safe completion was .7222 with paired seed/group 95% interval [.4722, .9167], and clear-time-only completion was .8194 [.7500, .9028]. Unnecessary stopping also appeared (.0278 and .0139, respectively). P1 was worse than P0 on those two changed-clearance strata. Thus the primary near-perfect result does not persist uniformly under new temporal values. Under the second paraphrase, P2 accuracy fell to .7917 despite .9583 safe completion.

Blanking the image changed P2 predictions in .2639 of cases; retained-reference accuracy was .6250 versus .6806 with the original image. Swapping the image changed .3056 of predictions and increased retained-reference accuracy to .8611, but accuracy against the image-implied rule was only .0972 and violation was .8750. The image affects behavior, yet these interventions do not support reliable visual grounding. Candidate/text shortcuts or misgrounded temporal cues remain plausible; the intervention does not identify a unique mechanism. Reporting only the improved retained-source score would conceal this failure.

The executable behavior test binds the model-selected candidate’s entry time and entry/holding

flag to the Core variables, together with source clearance. SAT/UNSAT is cross-checked against an independently implemented environment rule. The prediction is the object being tested, never the reference. Holding can be SAT without achieving route progress, which is scored separately. All 2,160 selected-action executions agreed with the independent source-rule evaluator; 216 additional swapped-image bindings were checked separately. This agreement validates the scoped behavior-test implementation, not the correctness of the model’s chosen action.

D4 compares the same 192 candidate records with verification bypassed or enabled: 24 valid controls and 24 records for each of seven injected error types. Unit errors are rejected at the schema layer; threshold changes are detected by source correspondence and independent Core boundary queries; comparison reversal and release deletion are caught by the Core/environment comparison; numeric and negation errors in CNL are caught by the supported grammar correspondence check. All 144 supported contract-error injections are detected, with zero false rejections among 24 valid controls. A separate negative control corrupts both the supplied source and its consistent contract: all 24 such errors are missed. Thus verified acceptance is 48/192 and valid-control acceptance is 24/24. These deliberately balanced injection counts are not natural error prevalence. Internal consistency alone accepts the wrong-but-consistent rules; source truth and whole-CoC factual correctness are outside this guarantee.

A separate expansion dataset reserves clearance times 6.0–9.9 seconds for training, 10.0–10.9 for validation, and 11.0–11.9 for test. Its 960/240/240 paired rows contain 40/10/10 distinct images and have no pixel overlap with D1–D4. These prepared data are not outcomes of expanded training. The authorized expanded comparison retains the original five controls and adds verified CNL; its actual checkpoint and evaluation results are reported separately when complete.

7 Limitations

The historical renderer generates only three distinct standard storyboards and three unseen-time storyboards. New generator seeds reuse those templates and candidate rotations; they are not fresh visual environments. The test exposes contract information at inference and cannot support a claim about inference without constraints. Supplied synthetic rules are not automatically extracted road facts. Solver satisfiability does not prove learned safety. Three model seeds do not replace independent environments. Negative or inconclusive preservation results remain reportable. Actual image tensors enter every model call, but this alone does not demonstrate visual grounding: the regular candidate time schedule can also reveal clearance timing. No image-ablation or independent natural-image transfer claim is made. The P1–P2 comparison evaluates preservation through the verified rendering pipeline. Verification’s contribution is scoped constraint-quality control, evaluated separately by D4; an additional causal gain in VLM performance is not the hypothesis. Wording and representation change together, and these results do not establish superiority of EBLC over every simpler representation. The comparison also changes constraint exposure jointly during training and inference. Actual optimizer updates are verified, but their causal contribution cannot be separated from inference-time conditioning without an additional matched pretrained control in this new protocol.

8 Real-scene case study

Existing single-reviewer development ACTION references, source confirmations, unknown-motion states, scoped exclusions, and disagreements are preserved. They are not consensus gold. The two accepted STOP cases and the separately reviewed scene18 launch check do not establish road-level

effectiveness. The frozen pretrained Qwen development baseline produced 19 valid predictions from 133 causal frames on 17 previously exposed clips, with no optimizer updates. Against the preserved scoped ACTION references, inappropriate selections were 10/16, normal-progress selections 2/8, and unnecessary stops 7/12. Their distinct denominators reflect unknown references and three clarified field masks. These appropriateness assessments are not certified collision or safety violations. The separate scene18 launch check saved one-update adapters for two conditions; it is execution evidence rather than efficacy evidence. No road review was repeated or generated to support the controlled temporal experiment.

9 Reproducibility

The owner of new code and results is Project04, **guardsynth-coc**. The paired protocol export pins the original Project01 world and trainer hashes, six original result records, the renderer, and the fixed protocol. Original runs remain immutable. New adapters, losses, raw per-example predictions, and output hashes are stored under the owner-scoped **paper1-method-preservation-001** artifact namespace. The run uses only local resources and the pinned Qwen revision:

89644892e4d85e24eaac8bacfd4f463576704203

No expanded LLM labels were needed for this supplied-rule task; consequently, no human sample-audit response or machine label is represented as human gold.